

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Summer Examination – 2023****Course: B. Tech.****Branch: CSE(AI&ML)****Semester: VI****Subject Code & Name: BTAIC601 / Deep Learning****Max Marks: 60****Date:****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Differentiate between Artificial Neural Network and Biological Neural Network.	(L4 / CO1)	6
B) Explain perceptron learning for ANN.	(L1 / CO1)	6
C) Compare and contrast linearly separable and linearly non separable problems with suitable example.	(L2 / CO1)	6
Q.2 Solve Any Two of the following.		12
A) Describe the architecture of Convolutional Neural Network.	(L1 / CO2)	6
B) In CNN what are the pros and cons of using max pooling layer Vs. Average pooling layer.	(L2 / CO2)	6
C) Enlist and justify the advantages of using Convolutional Neural Network Vs. Dense Neural Network for image classification.	(L5 / CO2)	6
Q. 3 Solve Any Two of the following.		12
A) What are the different ways of transfer learning? Explain them in brief.	(L1 / CO3)	6
B) Explain back propagation learning algorithm in detail.	(L1 / CO3)	6
C) Discuss the different techniques for handling overfitting issues during deep learning process.	(L2 / CO3)	6
Q.4 Solve Any Two of the following.		12
A) What is a Recurrent Neural Network (RNN)?	(L1 / CO4)	6
B) What is “Dead Neuron Problem”? How can Leaky ReLU overcome this problem?	(L2 / CO4)	6
C) Discuss in detail the implementation of AND gate using Mc Culloch Pitts neuron model.	(L2 / CO4)	6

Q. 5	Solve Any Two of the following.		12
A)	Explain and enlist the applications of Neural Style Transfer.	(L1 / CO5)	6
B)	How image denoising is achieved using Deep Learning techniques.	(L4 / CO5)	6
C)	Write Short note on Generative Adversarial Network.	(L2 / CO5)	6

***** End *****

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